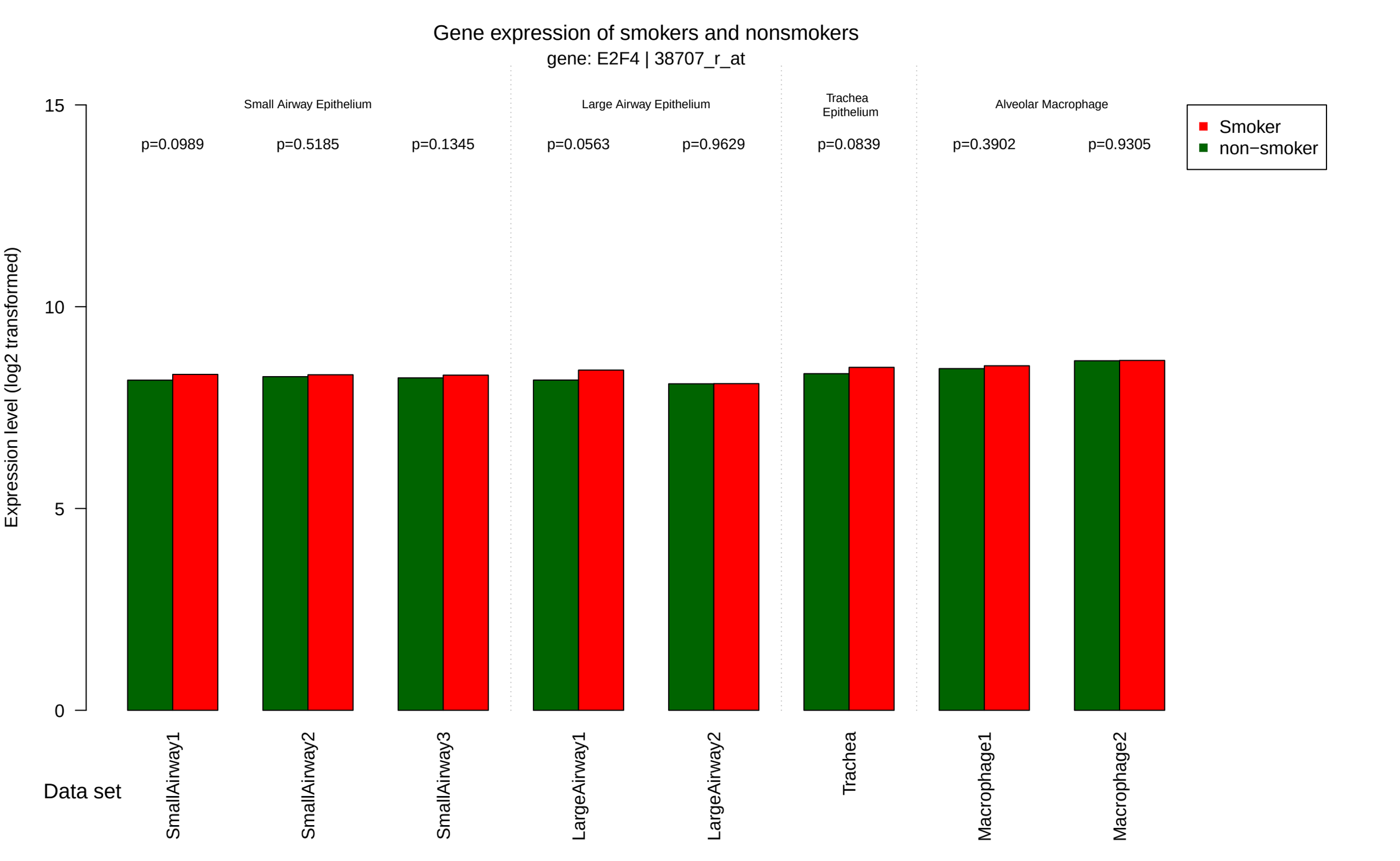
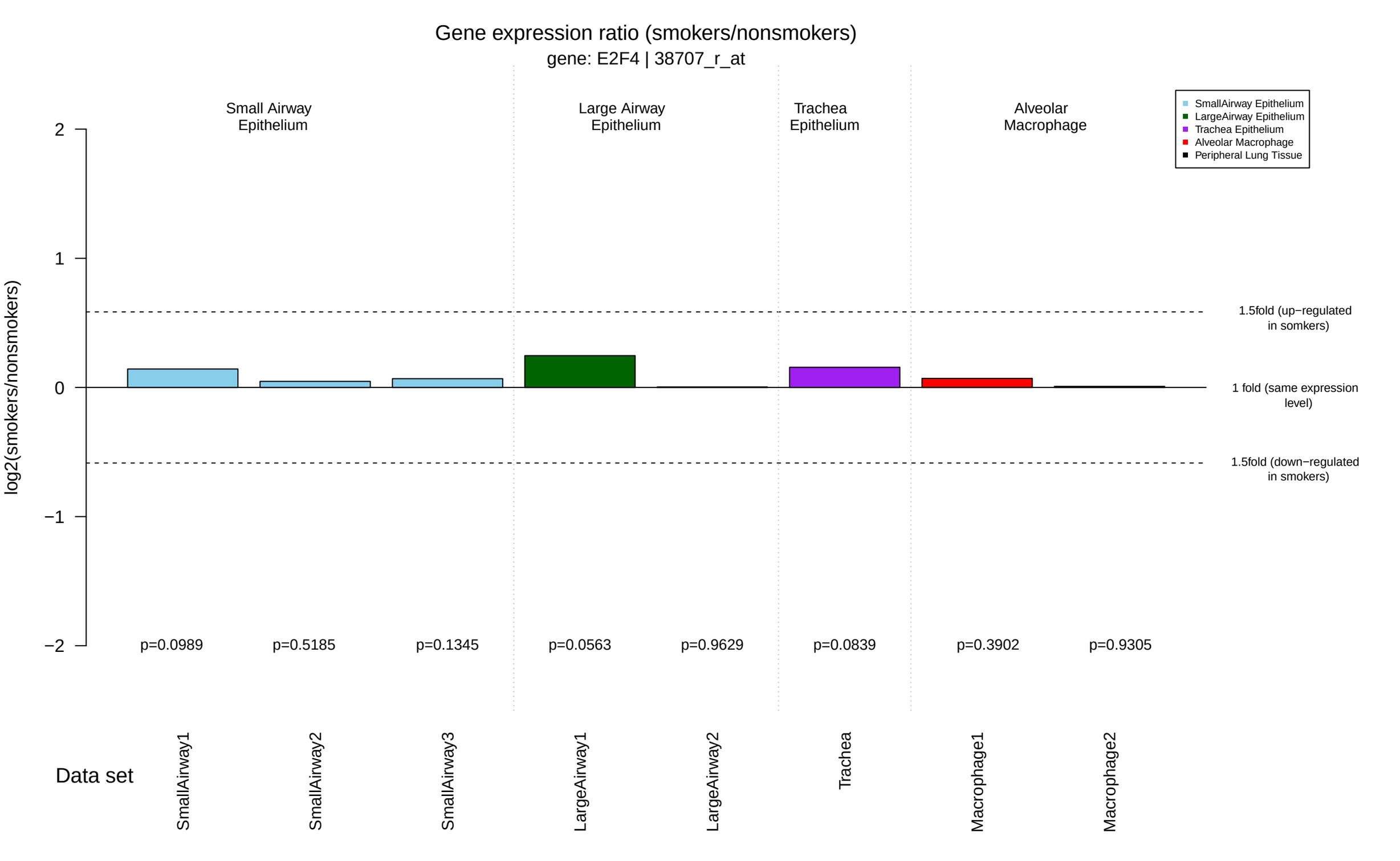
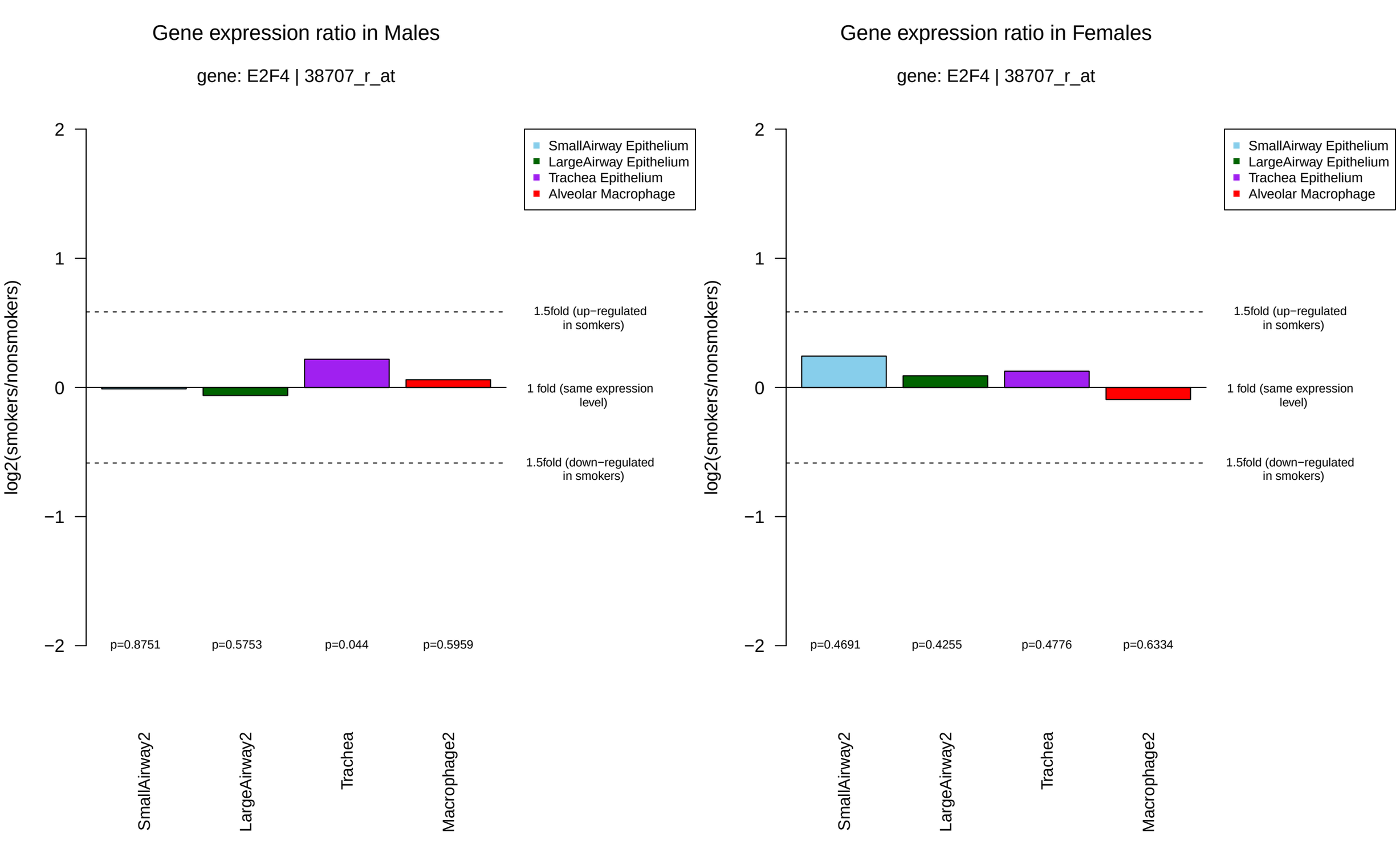
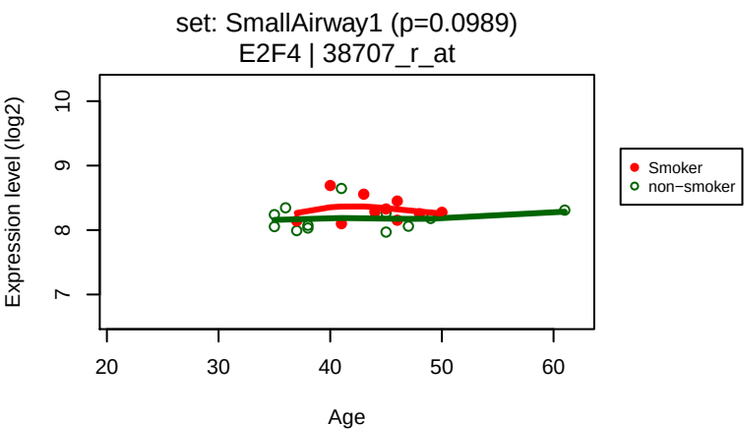
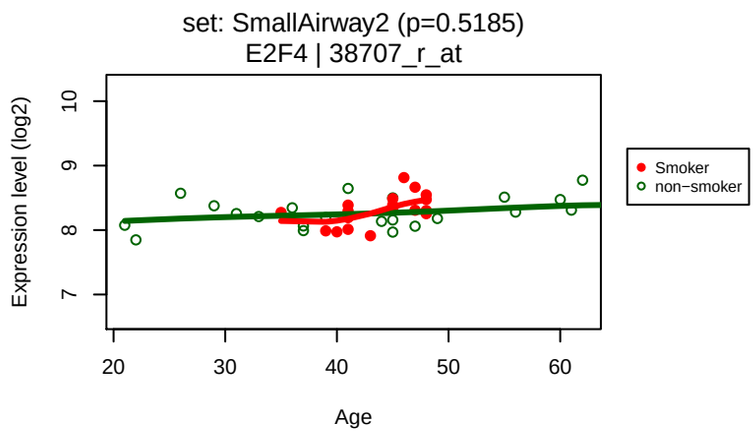
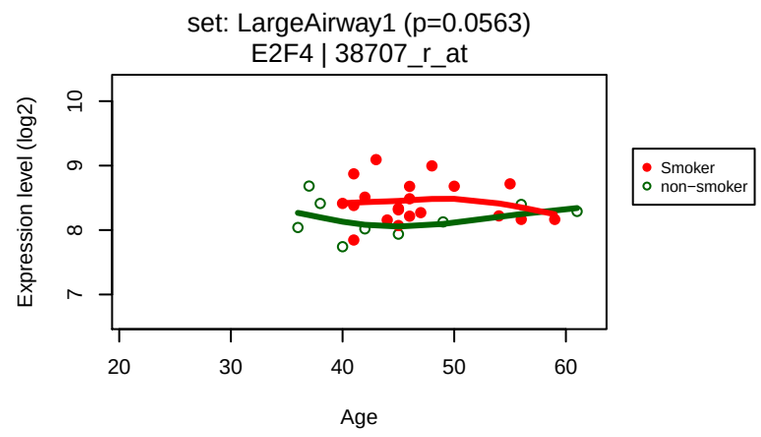

Gene expression of smokers and nonsmokers
gene: E2F4 | 38707_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0989
p=0.5185
p=0.1345
p=0.0563
p=0.9629
p=0.0839
p=0.3902
p=0.9305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

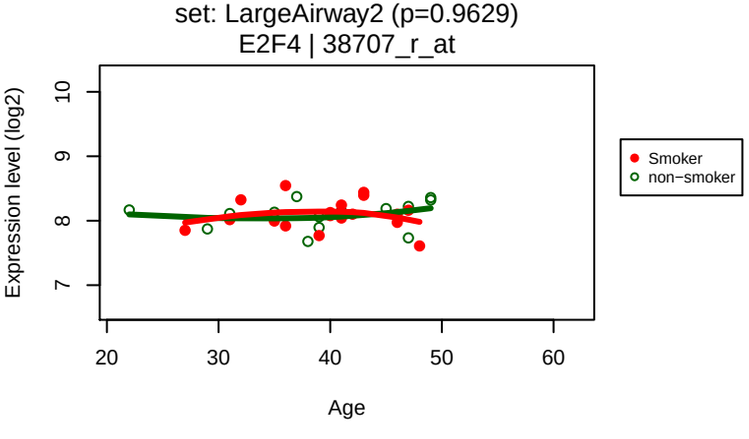

Gene expression ratio (smokers/nonsmokers)
gene: E2F4 | 38707_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0989
p=0.5185
p=0.1345
p=0.0563
p=0.9629
p=0.0839
p=0.3902
p=0.9305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

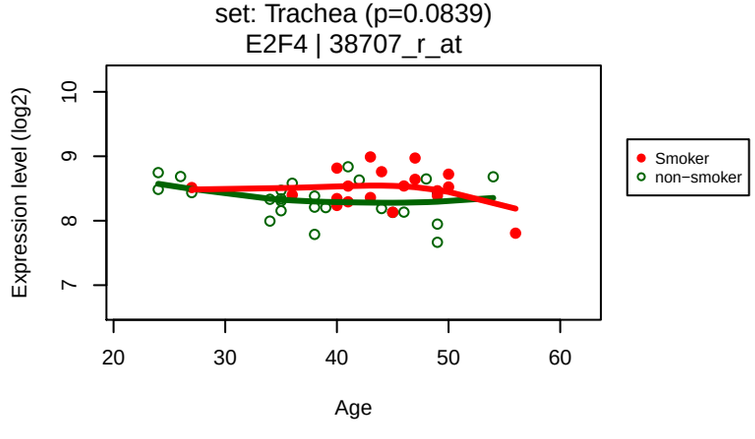

Gene expression ratio in Males
gene: E2F4 | 38707_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8751
p=0.5753
p=0.044
p=0.5959
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: E2F4 | 38707_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4691
p=0.4255
p=0.4776
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

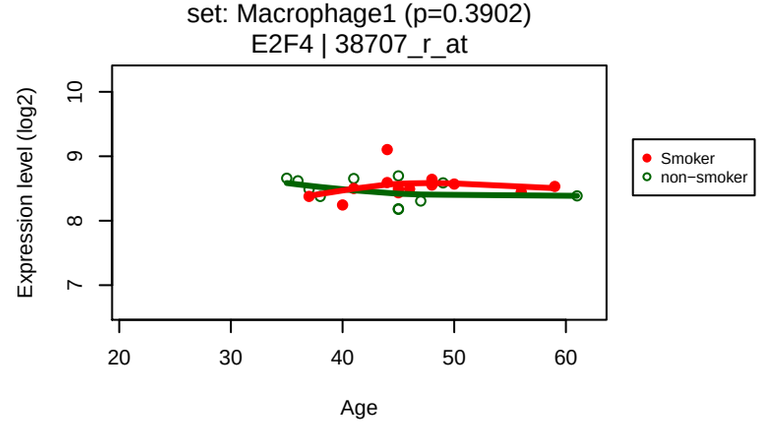

set: SmallAirway1 (p=0.0989)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker

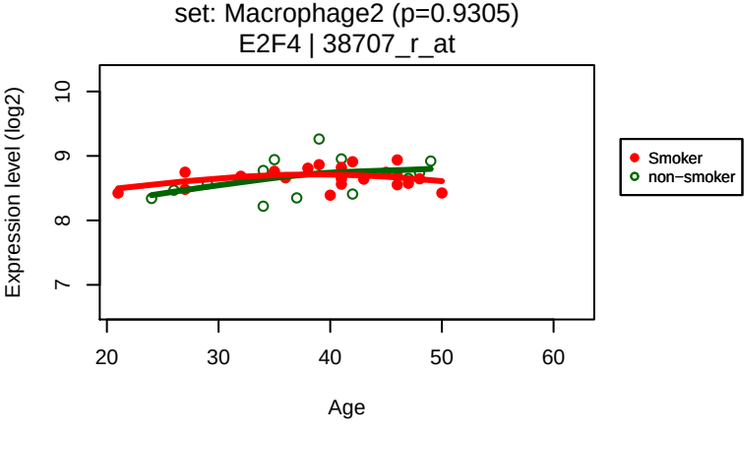

set: SmallAirway2 (p=0.5185)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker

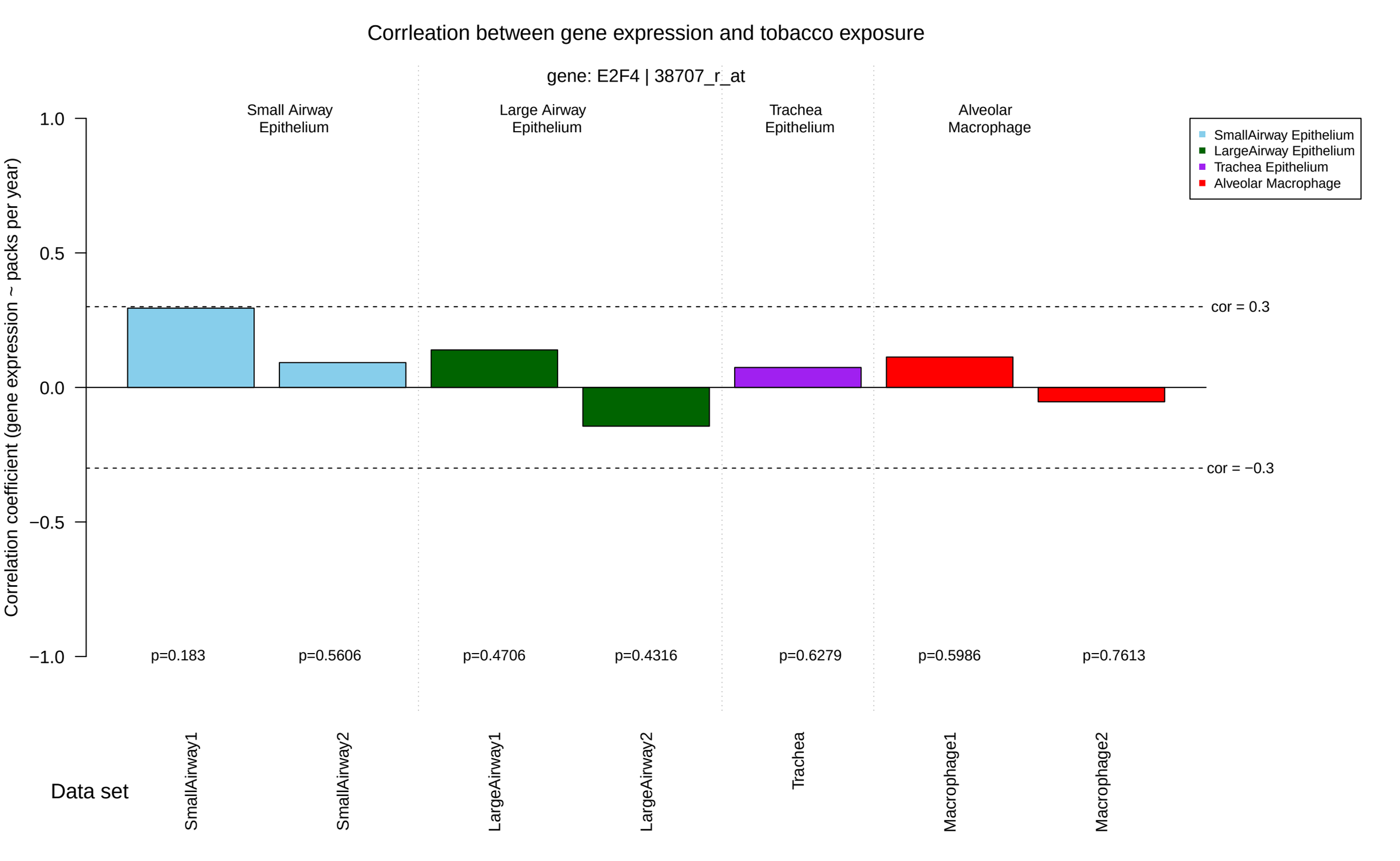

set: LargeAirway1 (p=0.0563)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9629)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0839)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3902)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9305)
E2F4 | 38707_r_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: E2F4 | 38707_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.183
p=0.5606
p=0.4706
p=0.4316
p=0.6279
p=0.5986
p=0.7613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2